Applying our learning

In this chapter, we learned how to create training sets with Databricks Feature Engineering, make baseline models using AutoML, track our experiments with MLFlow, explore Databricks Marketplace, and use the LLM Playground to enhance our data science projects. Now let's apply what we have learned to the datasets we cleaned in *Chapter 4* and featurized in *Chapter 5*.

Technical requirements

Here are the technical requirements needed to complete the hands-on examples in this chapter:

- We will integrate our LLM model with the **ChatGPT** model from **OpenAI**. You will need an API Key and need to install the OpenAI Python library.
- For the Multilabel Image Classification project work in this chapter, you will need a GPU cluster. Here is the single node GPU configuration.

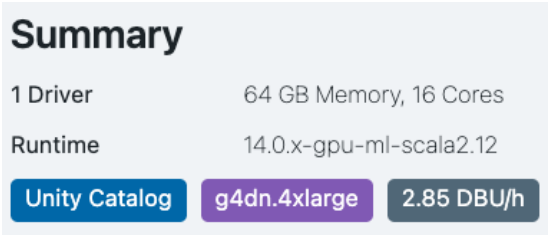


Figure 6.7 – Single node GPU cluster configuration used for this book

Project: Favorita Store Sales - Time Series Forecasting

Your goal is to create a `training_set` specification using DFE `FeatureLookups` and then create an AutoML baseline model. You will create the training set using the feature tables you created in *Chapter 5*. In *Chapter 4*, we used AutoML to jump-start our exploratory data analysis. Now, you will use AutoML to create a baseline model.

To follow along in your own workspace, please refer to the following notebook:

- `CH6-01-Building Favorita Baseline Model`

Building a training set

When you open `CH6-01-Building Favorita Baseline Model`, you will see that the repository code includes imports and variable names. It also includes a quick look at your data to remind you what the tables look like. Then we create the `model_feature_lookups` by specifying the `FeatureLookups` for each feature table, as shown in Figure 6.8. The `FeatureLookup` objects always include the `table_name` and `lookup_key`. Specifying the `feature_names` is optional.